TESLA ROADSTER THRUST & SUCTION SYSTEMS

SCHEMATIC · NOT TO SCALE · NOT A TESLA DRAWING

Tesla showed its next Roadster at a SpaceX test site in Texas on 15 October 2026. It drove up a ramp and hung upside down, flipped in the air, leapt over a row of Teslas and landed on jets of air. These sheets explain the physics behind each trick. Every figure is either from a named report or marked as my own estimate.

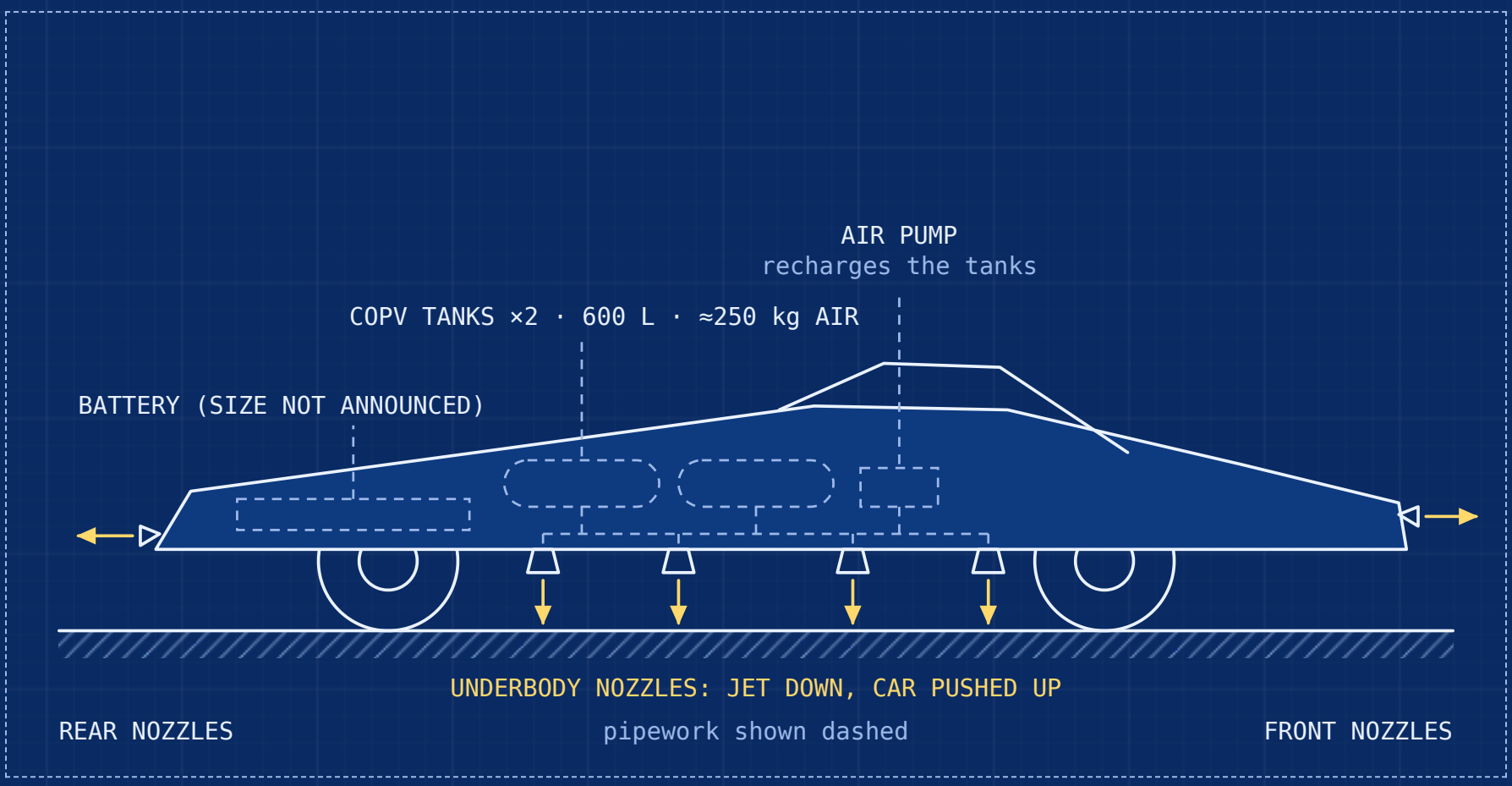
FOUR CORRECTIONS TO THE COMMON VERSION OF THE STORY

- **No jet engine and no fuel.** The thrusters use compressed air from tanks. Nothing burns.
- **It hops. It does not fly.** The demo car had no driver and was not shown in sustained flight.
- **The upside-down grip is suction fans.** It is not a suspension feature.
- **The Chinese tunnel loop is a different trick.** It relies on speed and a rigid chassis, and a Mercedes SLS AMG did a similar loop earlier.

WHAT WAS REPORTED

ITEM	REPORTED VALUE	STATUS
Air thrusters	10 nozzles on ultra-high-pressure air	Several outlets
Air storage	2 composite tanks, 600 L total, about 250 kg of air, refillable on board	Tesla engineer, 2 outlets
Suction fans	Up to 2 g of downforce, fitted to all three trims	Tesla engineer
0–60 mph	0.81 s in a video, first quoted as 1.0 s	Claim
Top speed	Over 230 mph (one outlet says over 250)	Conflicting
Range	300+ miles (two outlets) or 620 miles (one outlet)	Conflicting
Battery size	Not announced at the event	Not disclosed
Production	2028, per Musk	Promise

COMPRESSED-AIR THRUSTERS



HOW IT WORKS

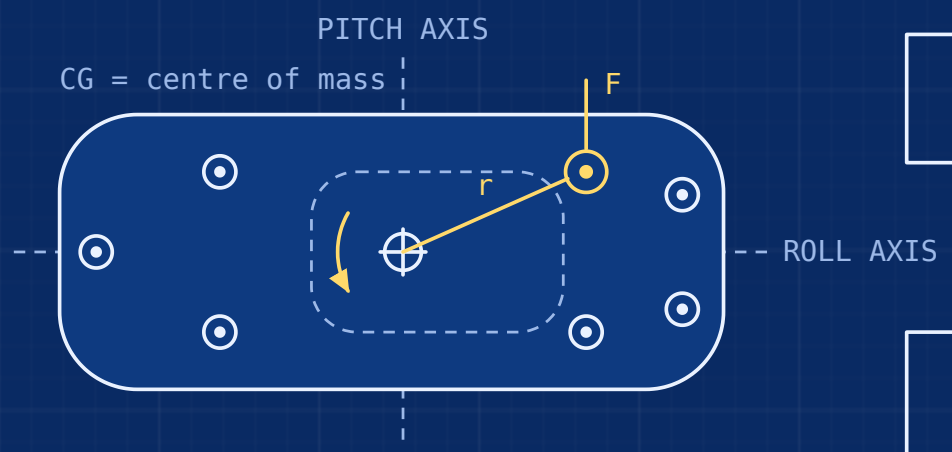
Air at very high pressure is released through a nozzle. The air rushes one way and pushes the car the other way. That is Newton's third law, the same principle SpaceX uses for small attitude jets. Nothing is burned, so there is no flame and no fuel tank.

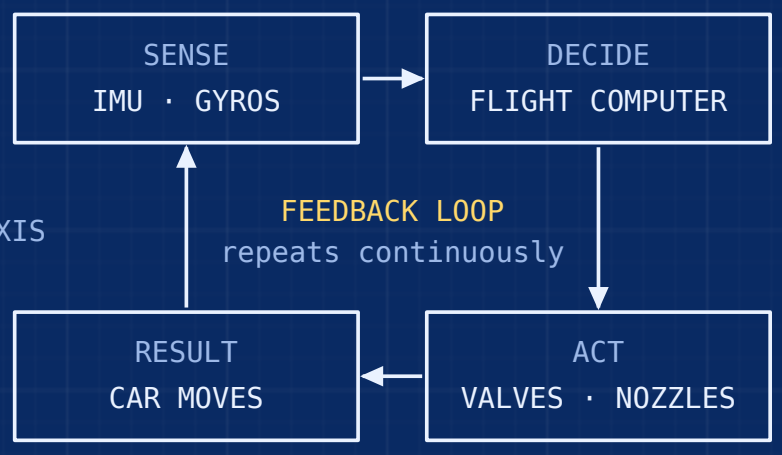
WHAT THE NUMBERS IMPLY (MY ESTIMATE)

- 250 kg in 600 L is about 417 kg per cubic metre, roughly 340 times denser than air at sea level.
- That is about 350–450 bar by the ideal gas law, with real-gas effects pushing it higher. Musk's 2018 plan quoted 10,000 psi (690 bar).
- Reports say 4 nozzles are under the car and 2 at the front. The positions of all 10 were not published.

SHEET	SOURCES	ESTIMATES	SCALE
2 · Air system	TechCrunch, Paul Tan, InsideEVs	Density and pressure	NTS

STAYING LEVEL IN THE AIR





⊙ force on car, toward you (jet points down)

Uneven thrust → torque τ = r × F → the car rolls or pitches

THE PROBLEM

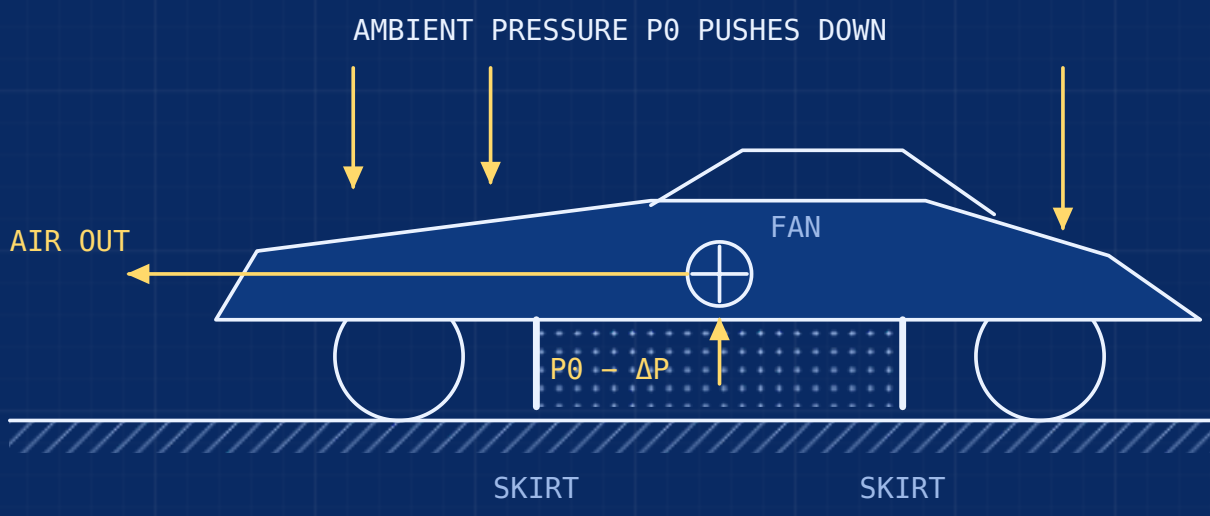
A car held up on jets has nothing like wings or wheels to keep it steady. If one jet pushes harder than the others, the car turns around its centre of mass. Left alone, a small tilt grows into a flip. Pushing at a distance **r** from the centre makes a turning force **τ = r × F**.

THE FIX

Sensors measure the tilt, a computer works out which nozzles to open and by how much, and the nozzles correct it. This loop runs again and again, many times a second. Rocket landings work the same way. In the demo, reports say two front thrusters helped turn the car right side up. Tesla has not published its control design, so this loop shows the standard method, not Tesla's actual code.

SHEET	SOURCES	ESTIMATES	SCALE
3 · Balance	InsideEVs (nozzle positions)	Loop is generic	NTS

FANS THAT PIN THE CAR DOWN

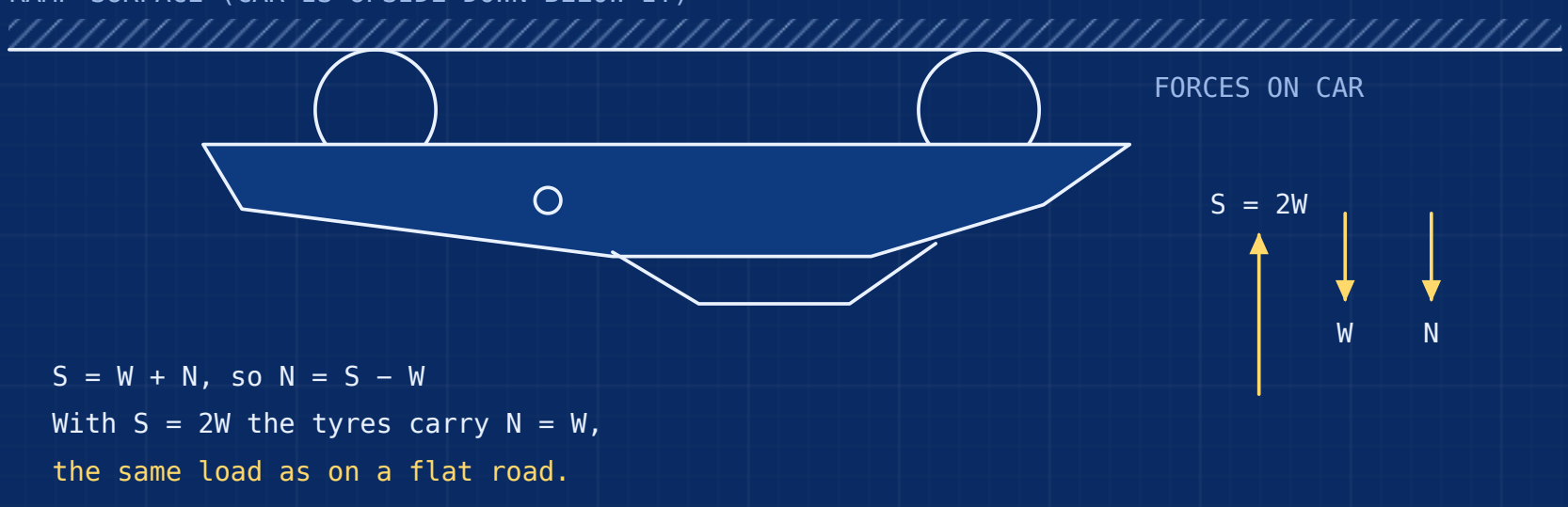


SUCTION FORCE

$F = \Delta P \times A$

2 g on 2,000 kg
= 39 kN

A = 2 m² → ~20 kPa
A = 4 m² → ~10 kPa



S = W + N, so N = S − W
With S = 2W the tyres carry N = W,
the same load as on a flat road.

HOW THE GRIP WORKS

Skirts seal a space under the car. Fans pull air out of that space, so the pressure there falls below the pressure outside. The outside air then pushes the car toward the road with a force of ΔP × A. For 2 g on a 2-tonne car, that is 39 kN. Over 2–4 m² of floor, the pressure drop is only about 0.1–0.2 of an atmosphere.

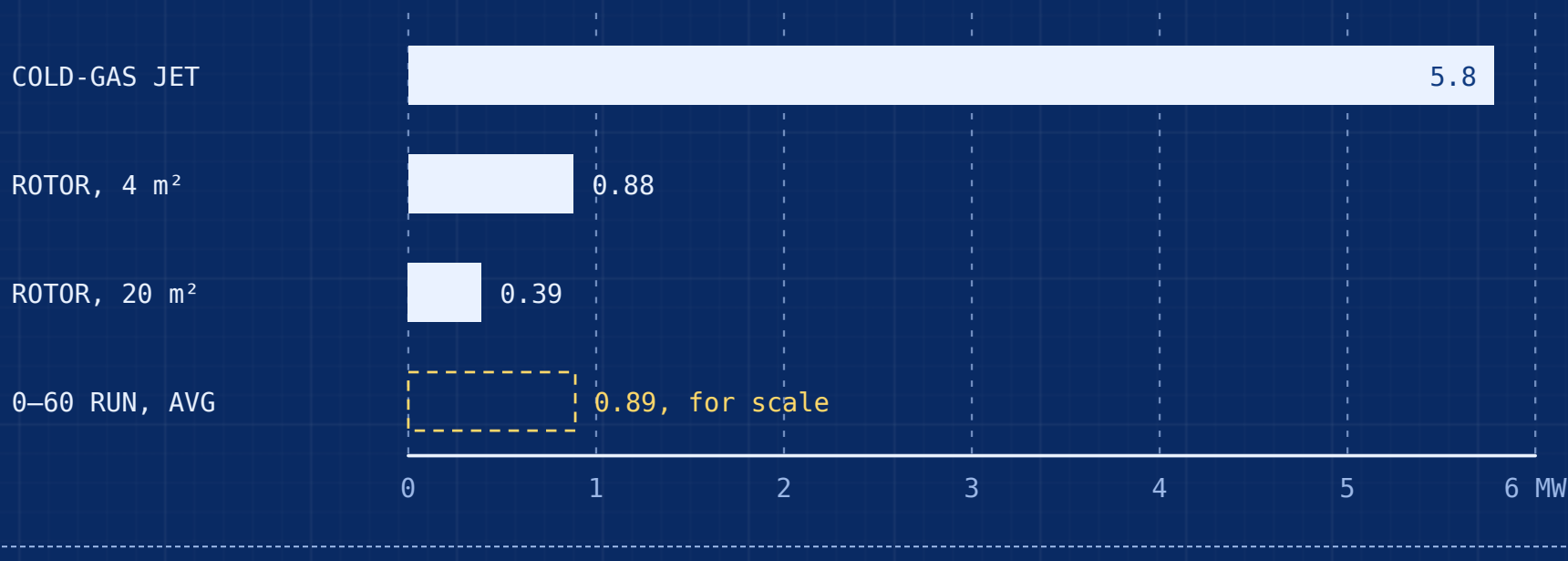
HISTORY AND LIMITS

Race engineers have used the same idea since the Chaparral 2J (1970) and the Brabham BT46B (1978). Reports compare the Roadster to the McMurtry fan car. Tesla says tyres alone give about 1.26 g of grip, and the fans let it corner above 2 g. Upside down, suction must exceed the car's own weight before the tyres press on the surface at all.

SHEET	SOURCES	ESTIMATES	SCALE
4 · Suction	Paul Tan, InsideEVs, Yahoo Finance	Area and ΔP, mass 2,000 kg	NTS

WHY IT HOPS AND DOES NOT CRUISE

I assumed a 2,000 kg car because Tesla gave no weight. All figures below scale with the real weight, and a heavier car makes every hover figure worse.



AIR EXIT SPEED (ASSUMED)	EXIT SPEED	AIR USED TO HOVER	HOVER TIME ON 250 KG
Low (specific impulse 50 s)	490 m/s	40 kg/s	≈ 6.7 s
Mid (60 s)	589 m/s	33 kg/s	≈ 8.0 s
High (70 s)	687 m/s	29 kg/s	≈ 9.3 s

READING THE CHART

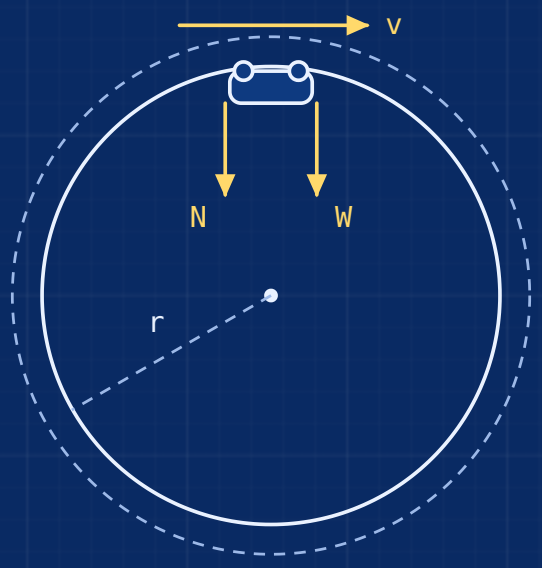
Holding 2 tonnes in the air needs about 20 kN of thrust. A real rotor moves a lot of air slowly, which is cheap in power. A jet of air at 500–700 m/s is the opposite, so it needs about 5.8 MW, more than six times what a 4 m² rotor needs. That is the reason helicopters and eVTOLs use big rotors.

REFILLING THE TANK

The pump takes its power from the battery. Squeezing 250 kg of air to about 360 bar takes at least 35 kWh even with perfect cooling, and real compressors need more. The speed ceiling for room-temperature air is about 780 m/s, so even a perfect nozzle cannot hover for much longer than ten seconds.

SHEET	METHOD	ESTIMATES	SCALE
5 · Energy	Rocket equation, momentum theory	All values on this sheet	Bars 90 px per MW

THE CHINESE TUNNEL LOOP



AT THE TOP OF THE LOOP

$N + W = m\ v^2 / r$

$N \geq 0$ means $v \geq \sqrt{(g\ r)}$

130 km/h = 36.1 m/s

$r \leq v^2 / g \approx 133$ m

Loop radius was not reported.
Any loop up to ~133 m radius stays on the wall by speed alone.

Reported apex speed loss: 3–5 km/h
Reported entry steering angle: 16°

WHAT WAS REPORTED

On 13 August 2026 Voyah said its Zhui Guang S became the first unmodified production car to complete a 360° loop in a closed test tunnel in Wuhan. It entered at about 130 km/h. The car has an aluminium double-wishbone front and five-link rear suspension, ten aero ducts, and 475 kW from two motors. The report says the Mercedes SLS AMG, a lighter supercar, had done a similar loop before.

WHAT IT IS NOT

The loop does not use suction. At speed the car stays on the wall because its path is curved, the same reason water stays in a swung bucket. Tyres and suspension then have to take the load and hold the car's line. It is a real engineering result, but it is separate from Tesla's fans and does not show that Mercedes failed at anything.

SHEET	SOURCE	ESTIMATES	SCALE
6 · Tunnel loop	Internet Info Agency, 13 Aug 2026	$r \leq v^2 / g$	NTS

FROM CARTOONS TO THE TEST PAD

1917	Curtiss Autoplane, an early attempt at a flying car.
1949	Taylor Aerocar, a road car that could be fitted with wings and flown.
1962	The Jetsons put a bubble-top flying car on television.
1970	Chaparral 2J, a race car that used fans to suck itself onto the track.
1974	The Man with the Golden Gun, a Bond film with a flying car.
1978	Brabham BT46B fan car, banned after one race.
1989	Back to the Future Part II, with a flying DeLorean.
Jun 2018	Musk says a SpaceX package for the Roadster will include about 10 small thrusters.
Aug 2025	Tesla is granted a patent for a fan suction system.
13 Aug 2026	Voyah completes the 360° tunnel loop in Wuhan.
15 Oct 2026	Tesla shows the Roadster with thrusters and fans in Texas.

Dates before 2018 come from general knowledge and were not checked in this session's searches.

WHAT IS STILL UNKNOWN

- Battery capacity, power output and weight. Range figures conflict between outlets.
- How long the car can stay airborne. The demo showed short hops with no driver.
- Whether a road-legal car can carry the tanks and still meet crash rules. Edmunds noted the demo car bounced as if it had no suspension.
- Whether 2028 production happens. A Smithsonian curator told CNN a hybrid would be a poor aircraft and a poor car.

SOURCES

CNN via ABC17: Tesla unveils what it says is a flying car

Edmunds: Tesla Roadster revealed

Paul Tan: tank and fan details

InsideEVs: the tricks the Roadster did

TechCrunch: Roadster with SpaceX thrusters

Interesting Engineering: flying car stunt

Internet Info Agency: Voyah tunnel loop

Fortune, 2018: SpaceX thruster package